Nick Leung EE446 SP26 Lab 9 Report

“hi” classification:

```
hi: 0.999961
sup: 0.000039
```

“sup” classification:

```
hi: 0.295928
sup: 0.704072
```

The main issue in the notebook was that the compile step set the optimizer, loss, and metrics for a model that’s more suited for regression. Since the data has two output classes, it makes more sense to change the optimizer to Adam, use binary cross-entropy loss (for two output classes), and use accuracy as the metric.

I also noticed the validation losses were increasing with each training epoch while the training losses decreased. This meant the model could be overfitting. To address this, I added L1 regularization to the hidden layers. I also decreased the hidden neurons to a layer of 32 and a layer of 16 to minimize the model size while keeping accuracy. This kept the validation loss and training loss a little closer together. I also updated the METRIC_TO_PLOT variable to “accuracy.” Model export stayed the same for me.